Deep-Research Report — Establishing a Defensible Research Gap for *Sistem Informasi Assessment Berbasis COBIT 2019*

> **Working title under review:** *Rancang Bangun Sistem Pengukuran Capability Level COBIT 2019 Berbasis Web dengan Traceability Evidensi Audit*
>
> **Date of report:** 2026-10-13 **All sources accessed:** 2026-10-13 (unless a different date is stated)
> **Repository:** `cobit-web` (`docs/06-deep-research-research-gap.md`) **Supersedes / reconciles with:** `01-literature-review.md`, `02-system-design-research.md`, `03-thesis-decision-log.md`, `04-regulasi-indonesia-cobit.md`, `05-product-requirements-document.md`, `latar-belakang.md`

This report is **adversarial to the existing drafts on purpose**. It independently re-checks the claims that the thesis's novelty rests on, separates verified fact from interpretation, labels every claim with its evidence status, and flags where the current drafts over-claim. It does **not** invent sources or convert "not found" into "does not exist."

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1. Executive Summary

Overall verdict: a defensible gap exists — but it is *not* "Excel → web".

The research gap is defensible only when framed as an **Information Systems design problem**:

> *ISACA provides no official Process Assessment Model (PAM) or computational assessment instrument for COBIT 2019. The only official digital artifact — the COBIT 2019 Design Guide Tool Kit — is a **design/prioritisation spreadsheet**, not a capability-measurement engine. The published COBIT 2019 rating scale (F/L/P/N) is defined as an assessor **judgment**, and ISACA/ISACA-Journal sources give "no single mandatory aggregation formula". Existing commercial GRC tools map COBIT content but do not compute COBIT capability levels; academic prototypes move the spreadsheet to the web without documented scoring or evidence traceability. There is therefore a documented, reproducible gap for a system that (a) makes the CPM scoring/aggregation logic **explicit and inspectable**, (b) links every activity rating to **audit evidence** (traceability), and (c) manages the assessment as a **repeatable, comparable, reportable IS artifact** (multi-assessment, history, comparison, reporting).*

1.1 Key findings (with confidence)

#	Finding	Confidence	Anchor
F1	**COBIT 2019 has no PAM.** COBIT 5 did (ISO/IEC 15504/330xx); COBIT 2019 moved to a CMMI-based scheme called COBIT Performance Management (CPM). ISACA's own text: "*No*	**High**	*COBIT 2019 Framework: Introduction and Methodology* §6.3 (p. 38); ISACA Journal 2021 Vol 6 (Gorgona)

	separate process assessment model (PAM) publications are necessary, nor will they be provided with COBIT 2019."		
F2	**The official digital toolkit is design-only.** The Design Guide Tool Kit weights the 40 objectives by design factors to produce a prioritised design (initial/refined/concluded scope). It is **not** a capability-assessment instrument.	**High**	ISACA Industry News 2020 (Harisaiprasad); ISACA 2020 "COBIT Tool Kit Enhancements"
F3	**The N/P/L/F thresholds used by the prototype are correct and match ISO/IEC 33020:2019 §5.3 exactly** (N 0–≤15%; P >15–≤50%; L >50–≤85%; F >85–≤100%). COBIT 2019's own prose is loose at the 15/50/85 boundaries; ISO resolves them.	**High**	*COBIT 2019 Framework: I&M* §6.4.2 (p. 39); ISACA COBIT-Focus 2019 (Dhulipalla); **ISO/IEC 33020:2019 §5.3**
F4	**ISACA defines *no* mandatory aggregation formula.** Simple average is proposed for early maturity; weighted average is "recommended" otherwise. The 2021 Journal article explicitly says organisations must define the **expected evidence per activity**.	**High**	ISACA Journal 2021 Vol 6
F5	**The rating scale is a judgment, not a mechanical count.** ISACA: "*Fully … remains a judgment call, but it can be substantiated by the examination or assessment of the components of the enabler.*"	**High**	ISACA COBIT-Focus 2019 / Souza Neto et al. 2019
F6	**There is no single official "gate"; three readings compete.** (i) COBIT's literal sentence: a process reaches a level "*as soon as **all activities of that level are performed successfully***"; (ii) the published **proposal** (Souza Neto, Almeida & Mira da Silva 2019): all activities **L or F**; (iii) the **ISO/IEC 33020 staged model**: the level *claimed* needs only **Largely (>50%)**, all lower levels need **Fully (>85%)**.	**High**	*COBIT 2019 Framework: G&M Objectives* §3.4 (p. 20); ISACA Industry News 2019; ISO/IEC 33020 (via Automotive SPICE PAM 3.1, Table 16)
	"Excel → web" by itself is an engineering activity, not a		

F7	**research contribution.** It becomes a gap only when tied to a validated problem (transparency, traceability, reproducibility, assessment management).	**High (interpretation)**	See §3
F8	**The RAG/AI layer is not needed to defend the gap.** The existing RAG-heavy framing (a) is not required by CPM, (b) contains at least three source-integrity problems (see §7), and (c) dilutes the IS contribution.	**Medium-High**	See §7
F9	**The prototype's "100% match vs Excel" test claim is not supported by the test that exists.** `engine.test.ts` compares against hand-coded literals and an **interview-derived demo dataset** (`benchmarkData.ts`; not a validated or official benchmark) — not against the ISACA toolkit (which is design-only, §5.1) and not against any official reference.	**High**	`src/utils/__tests__/engine.test.ts`; `src/data/benchmarkData.ts`
F10	**The prototype's gate (>85% mean at every level) is stricter than the ISO staged rule and a different operationalisation from COBIT's literal "all activities" sentence** — defensible only if labelled an *adaptation* and made configurable (see §4.2 B5, §4.3). It is recognisable in prior Indonesian practice (Marcela & Fianty 2023; Simatupang & Fianty 2023).	**High**	§4.2; *COBIT 2019 Framework: G&M* §3.4; ISO/IEC 33020:2019; DOI 10.33022/ijcs.v12i4.3353; DOI 10.33379/gtech.v7i4.3141
F11	**Indonesian regulation does not mandate COBIT, and is context — not the justification.** COBIT appears only as an optional best-practice reference, mostly in *revoked* instruments. Current SPBE-evaluation law is **PermenPANRB 8/2026** (revoking 59/2020); Perpres 132/2022 is *not* an amendment to Perpres 95/2018.	**High**	§9.5 (verified 2026-10-13)

1.2 What must change in the existing drafts (top 6)

1. **Stop claiming a COBIT 2019 PAM exists** — the drafts mostly get this right; keep it, and never let it drift into "COBIT 2019 assesses process attributes/outcomes like COBIT 5 PAM did."
2. **Downgrade "verified from COBIT 2019 I&M"** for the capability-level *names* in `01-literature-review.md` §3.4 — the framework text provides **six descriptive characteristics**, not the CMMI one-word labels; the drafts mixed two naming systems (see §4.4).
3. **Fix the "critical rule"** in `01-literature-review.md` §3.5/§3.3: the published gate is "*all activities rated **L or F***", not "*all activities rated **F***". The prototype (>85% aggregate) is a *different* rule (see §4.2).
4. **Correct the citation "Almeida 2019" → Souza Neto, Almeida & Mira da Silva (2019)**, ISACA Industry News (a *proposal*), not an ISACA standard.
5. **Remove or re-label the "validated 100% vs ISACA Excel" claim.** There is no ISACA Excel assessment workbook to validate against; the reference is the documented CPM rules or the PANRB workbook, both of which must be labelled as such.
6. **Rebalance the literature review away from RAG.** Move AI/RAG to an *optional future-work* paragraph; the gap must stand on the assessment/evidence/management argument (§7).

1.3 AI/RAG recommendation (short form)

OPTIONAL-SIMPLE — a single direct LLM call that drafts a recommendation from the gap result, always labelled "*AI-generated draft — assessor review required*". **No RAG pipeline, no retrieval metrics, no AI in the title, RQs, or methodology.** Rationale and the decision rule are in §7. (If time pressure is real, **OMIT** entirely — the gap does not need it.)

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2. Method and Evidence-Labeling Conventions

2.1 What was done

- Independently re-verified the **official COBIT 2019 / ISACA** primary sources (CPM, capability levels, rating scale, toolkit purpose, counts).
- Re-read the prototype's actual implementation (`src/utils/cpmEngine.ts`, `src/utils/__tests__/engine.test.ts`) and the PRD's CPM section, and checked them against the primary sources.
- Re-checked a sample of the RAG/audit claims in `01-literature-review.md` against the real literature.
- Cross-checked DOI-level references for DSRM and evaluation methods.

2.2 Evidence labels (used throughout this report and required in the thesis)

Label	Meaning
`[VER-P]`	**Verified – primary.** ISACA/ISO/IIA/peer-reviewed page fetched this session; the claim is visible on that page.
`[VER-S]`	**Verified – secondary.** Real, findable source, but the specific claim is reported by a third party, not the origin page.
`[2ND-UNV]`	**Secondary, unverified.** A number/name attributed to a source that could not be confirmed from the source page. Treat as suspect.
`[NF]`	**Not found in this search.** *Not* evidence of absence.
`[OWN]`	**Interpretation / reasoning by this report.** Not a source claim.

2.3 Known limitations

- **Paywalled primaries:** ISO/IEC 33020:2015/2019 §5.6 and ISO/IEC 33004:2015 body text are restricted; the COBIT 2019 *books* and the Design Guide Tool Kit were read this session from **publicly available PDF/ZIP copies** (content matches ISACA's published titles/archived pages), so exact quotes carry `[VER-P]` but should be cited via the ISACA store/ISBN, not a mirror. The ISO staged rule (§5.6) is quoted via the 2015 model reproduced in Automotive SPICE PAM 3.1 (Table 16).
- **Search instability:** the web-search tool failed intermittently; Brave Search via fetch was used, with occasional HTTP 429. `lite.duckduckgo.com` gave a certificate error. Where a search failed, the item is marked `[NF]` rather than assumed absent.
- **Vendor claims** (GRC platforms) were not all re-verified this session; those rows are labelled `[2ND-UNV]` and must be re-verified before they appear in the thesis.

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3. Focus A — Is "Excel → Web Migration" Itself a Valid Research Gap?

3.1 Direct answer

No — not on its own. Migration of a spreadsheet artefact to a web application is an *implementation activity* (a software engineering task). A thesis contribution (DSRM "design knowledge") requires evidence of a **problem class** that the artefact addresses, plus a **design principle** that generalises beyond the one artefact (Gregor & Hevner 2013; Hevner et al. 2004). "We put the Excel in the browser" satisfies neither.

It becomes a **valid** gap when the following chain holds, and each link is evidenced:

> (1) an authoritative artefact exists that is **only** a file → (2) it is **not** an assessment instrument → (3) the assessment task therefore relies on **un-inspectable, undocumented, non-traceable** practice → (4) existing tools/prototypes do **not** close this → (5) the thesis contributes a **system + explicit design principles** for the scoring/evidence/management problem.

3.2 The chain, item by item

Element	Status	Evidence
The official digital artefact is a **design** workbook, not an assessment engine	`[VER-P]`	ISACA Industry News 2020 (Harisaiprasad, 27 Apr 2020): the tool kit "provides a chart for weighting the 40 processes … initial/refined/concluded scopes"; ISACA 2020 "COBIT Tool Kit Enhancements"
No official COBIT 2019 PAM / capability-measurement instrument	`[VER-P]`	ISACA Journal 2021 Vol 6 (Gorgona): "*There is no PAM for COBIT 2019*"
COBIT 2019 capability ratings are designed to be **judgment + evidence based**	`[VER-P]`	ISACA COBIT-Focus 2019 (Dhulipalla): "Fully … remains a judgment call"; "traceability must be maintained between the objective evidence collected and the process activity ratings assigned"
Existing academic prototypes migrate the spreadsheet **without documenting scoring or evidence linkage**	`[2ND-UNV]`	Claims carried in `02-system-design-research.md` §2.5 / `03-thesis-decision-log.md` §3.4 — **must be re-verified from each paper before the thesis states it**
The thesis's own product decision already concedes the framing risk	`[VER-P]` (repo)	`05-product-requirements-document.md` line 78 and 43; `03-thesis-decision-log.md` line 38

3.3 How to phrase the gap (defensible vs indefensible)

❌ Indefensible	✅ Defensible
"No web version of the COBIT 2019 Excel tool exists."	"The official COBIT 2019 artefact is a *design* spreadsheet; no official *capability-assessment* instrument exists (no PAM), so scoring and evidence-linkage practice is undocumented and non-reproducible."
"We move the 27-sheet workbook to the web."	"We make the **CPM scoring/aggregation logic explicit and inspectable**, and bind each activity rating to **auditable evidence**, so a third party can reproduce the result (cf. IIA 2024 documentation principle; ISACA ITAF evidence requirement)."
"Faster data entry."	"Assessment **management**: repeated assessments across periods/units, historical comparison, and report generation — an IS artefact class, not a data-entry convenience."

3.4 Focus-A gap table

Element	Claim to be defended	Evidence anchor	Status	Verdict
A1 Official artefact is a file	Toolkit is `.xlsx`, distributed as a file	ISACA tool kit pages	`[VER-S]`	Supports "file-only"
A2 Artefact purpose	Design/prioritisation, not assessment	ISACA Industry News 2020	`[VER-P]`	**Core support**
A3 Assessment-specific instrument	None published for COBIT 2019	ISACA Journal 2021	`[VER-P]`	**Core support**
A4 Manual practice is non-reproducible	Scoring/aggregation not standardised	ISACA gives no mandatory formula	`[VER-P]`	Supports
A5 Prior prototypes do not close it	Undocumented scoring, no evidence mgmt	Draft claims	`[2ND-UNV]`	**Must re-verify per paper**
A6 Migration alone is a contribution	—	—	`[OWN]`	**Rejected**

Focus-A conclusion: the *migration* is not the gap; the **absence of an official, transparent, evidence-traceable COBIT 2019 capability-assessment instrument and its management workflow** is.

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4. Focus B — Validating the Official COBIT 2019 CPM Mechanism

> This is the section that most affects whether the thesis's scoring is defensible. The instruction was: **do not assume all scores can be averaged.** The evidence below shows *why*.

4.1 What ISACA actually specifies

(a) CPM is CMMI-based; COBIT 5's ISO/IEC-330xx PAM was dropped.

	COBIT 5	COBIT 2019
Performance model	ISO/IEC 33000 (SPICE)	CMMI Performance Management scheme
Scale	0–5	0–5 (capability *and* maturity)
PAM	Yes (ISO/IEC 15504/330xx)	**No**

Processes	37	40

[VER-P] *COBIT 2019 Framework: Introduction and Methodology* §6.3/§6.4.1 (p. 38): "*COBIT 2019 supports a CMMI-based process capability scheme … ranging from 0 to 5*"; "*In COBIT 2019, the explicit process outcomes or process goals are replaced by the process practices themselves*"; and, decisively, "***No separate process assessment model (PAM) publications are necessary, nor will they be provided with COBIT 2019***." The same point is official in ISACA (2018) *Introducing COBIT 2019: Major Differences with COBIT 5*. Corroborated by ISACA Industry News 2020 (Harisaiprasad) and ISACA Journal 2021 Vol 6 (Gorgona).

(b) Two distinct constructs — do not conflate them.

- **Capability level** — per **process**; "a measure of how well a process is implemented and performing."
- **Maturity level** — per **focus area**; "achieved if all required capability levels are achieved" across the processes in that focus area.

[VER-P] ISACA Industry News 2020 (Elue, "Effective Capability and Maturity Assessment Using COBIT 2019"); ISACA COBIT-Focus 2019 (Dhulipalla).

(c) Capability-level characteristics 0–5 (quoted from *COBIT 2019 Framework: Introduction and Methodology*, fig. 6.2):

Level	Characteristic (framework wording)
0	Process lacks basic capability; incomplete approach to the governance/management purpose; may or may not be meeting the intent of any process practices
1	Process more or less achieves its purpose through an **incomplete** set of activities — initial/intuitive, not very organised
2	Achieves its purpose through a **basic, yet complete, set of activities**; characterised as *performed*
3	Achieves its purpose in a much more **organised** way using organisational assets; processes typically **well defined**
4	Achieves its purpose, well defined, **performance quantitatively measured**
5	Achieves its purpose, well defined, measured to **improve** performance, **continuous improvement** pursued

[VER-P] Souza Neto, Almeida & Mira da Silva (2019), ISACA Industry News, quoting *COBIT 2019 Framework: I&M* fig. 6.2.

(d) The rating scale (F/L/P/N) — exact wording.

> **Fully (F)** — the capability level is achieved for **more than 85%**. *(This remains a judgment call, but it can be substantiated by the examination or assessment of the components of the enabler, such as process activities, process goals or organisational structure good practices.)* **Largely (L)** — achieved **between 50% and 85%**. **Partially (P)** — achieved **between 15% and 50%**. **Not (N)** — achieved **less than 15%**.

[VER-P] ISACA COBIT-Focus 2019 (Dhulipalla, 16 Sep 2019); reproduced by Souza Neto et al. 2019, who cite ISO/IEC 33004:2015 "as referenced in ISACA, *COBIT 2019 Framework: Introduction and Methodology*".

> **Boundary note:** ISACA's own prose is **ambiguous at the boundaries** (Largely "between 50 and 85"; Partially "between 15 and 50"; exact-15% sits only in P). The exact partition is resolved by **ISO/IEC 33020:2019 §5.3** — N 0–≤15%, P >15–≤50%, L >50–≤85%, F >85–≤100% — which is **identical to the prototype's ≤15/≤50/≤85/>85 mapping** (see B2). State in the thesis that the

percentages are cited from ISO/IEC 33020:2019, with COBIT's §6.4.2 prose as corroboration — not the reverse.

(e) The rating is per activity *per capability level*, and it is a judgment.

ISACA: "For each objective assessed, a rating is assigned for each process activity **up to and including the highest capability level defined in the assessment scope**. The rating is based on validated data, and **traceability must be maintained** between the objective evidence collected and the process activity ratings assigned." [VER-P]

(f) There is no official aggregation formula.

- *COBIT 2019 Framework: I&M* gives the scale but **no mandatory arithmetic** to roll activities → practices → objectives → domains.
- ISACA Journal 2021 Vol 6 (Gorgona) *proposes*: "For organisations in early maturity stages, a **simple average** calculation for the activity values can be used … If an organisation has a greater capacity … then a **weighted average** … is recommended." It also requires organisations to "**create milestones** … to identify the **expected evidence** for the capability level in each activity." [VER-P]
- Counts used by the framework: **1,202 activities → 231 practices → 40 objectives → 5 domains**. [VER-P] (Gorgona 2021; ISACA 2020 "COBIT Tool Kit Enhancements" confirms 231 practices).

(g) The "gate" that decides when a level is achieved — three competing readings.

Reading	Rule	Source	Nature
COBIT 2019 literal	"*A process reaches a certain capability level as soon as* ***all activities of that level are performed successfully***"	*COBIT 2019 Framework: G&M Objectives* §3.4 (p. 20)	**Official text** — effectively a 100% rule
ISACA-published proposal	A level is at least attended if **all** its activities are rated **L or F**; any **N/P** drops it	Souza Neto, Almeida & Mira da Silva (2019), ISACA Industry News	**Proposal**, not a standard
ISO/IEC 33020 staged model	The level **claimed** needs only **Largely (>50%)**; all **lower** levels need **Fully (>85%)**	ISO/IEC 33020 (quoted via Automotive SPICE PAM 3.1, Table 16)	**ISO measurement framework** that COBIT 5 inherited; COBIT 2019 does **not** bind to it

[VER-P] for all three. Souza Neto et al. also note processes influence each other via outputs ("garbage in, garbage out"), so they argue a low-scoring process feeding a high-scoring one should not be auto-capped.

4.2 Is the prototype's algorithm valid? (`src/utils/cpmEngine.ts`)

The engine implements:

```
score: Yes=1.0, Partially=0.5, No=0.0, N.A.=excluded
P_L   = sum(scores of level-L activities) / count(non-N.A. level-L activities)
rating: P_L ≤0.15 → N; ≤0.50 → P; ≤0.85 → L; >0.85 → F
gate : level passes iff P_L > 0.85   (i.e. "Fully")
base : level 1 assumed achieved; levels 2..5 evaluated in order; first failure closes the gat
gap  : max(0, target − currentLevel)
```

ID	Element	Assessment	Severity
B1	Threshold values match	□ F/L/P/N = >85 / 50–85 / 15–50 / <15	—

B1	ISACA	□ F/L/P/N = >85 / 50–85 / 15–50 / <15	—
B2	Boundary partition	□ **The engine's partition (≤15→N, >15–≤50→P, >50–≤85→L, >85→F) matches ISO/IEC 33020:2019 §5.3 exactly.** COBIT's own prose is loose at 15/50/85 (e.g. "between 50 and 85"); adopting the ISO boundaries is the correct disambiguation — cite ISO for the boundaries, not COBIT.	—
B3	Activities → percentage → band	□□ **Category shift.** ISACA assigns F/L/P/N **per activity** (judgment of how far the capability level is achieved). The engine derives a **percentage of activity-answers** and applies the band to *that*. Defensible as an operationalisation, but it is **not** ISACA's mechanism.	**High (must be disclosed)**
B4	Simple mean	□ Permitted for early maturity by Gorgona 2021 — *as a proposal*. Not officially mandated.	Medium (disclose)
B5	Gate ">85% aggregate = pass"	□ **Addressed (2026-10-13).** The gate is now **configurable and user-visible**: `aggregate-85` (default, preserves legacy behaviour; cite the published "cannot continue below 85%" precedents — Marcela & Fianty 2023; Simatupang & Fianty 2023), `all-l-or-f` (§4.1g published proposal), and `iso-staged` (§4.1g ISO staged model). The results page exposes a rule-sensitivity panel showing the level produced by each gate for the same data, and the method card states the rule is an *operational adaptation*, not an ISACA standard.	Closed (was High)
B6	3-point response scale (Yes/Partially/No)	□□ Drops ISACA's **L (Largely)** at the activity level; recovers it only after aggregation. Acceptable simplification, must be stated.	Medium
B7	`N.A.` excluded from denominator	□ **Addressed (2026-10-13).** `N.A.` is still excluded from the denominator, but a level whose activities are **all** `N.A.` is now flagged **Insufficient Evidence** (not N, and it halts the climb without asserting a rating). The status is surfaced in the aggregation trail.	Closed (was High)
B8	Level 1 assumed achieved	□ Reasonable for target-based assessment of selected objectives; but level 0 (process absent) cannot be produced. State the assumption.	Low–Medium
B9	Level with **no defined activities**	□□ `validCount = 0 → 0 → fail`, which *happens* to yield the intended "cap at level 4" outcome for level-5-less processes — but only **accidentally**; it is fragile and undocumented.	Medium
B10	No persistence / single assessment	□ **Addressed (2026-10-13).** Multi-assessment history + compact snapshots (`cobit_assessment_snapshots_v1`) + a baseline→candidate comparison page are now implemented (`AssessmentContext`, `CompareAssessments.tsx`).	Closed
		□ **Addressed (2026-10-13).** The claim is removed from `README.md`, `05-PRD`, and the test banner, and a genuine **independent reference-calculation**	

B11	"100% match vs Excel" claim	**oracle** (`referenceCalculation.test.ts`) now derives expected results by hand from the documented rules and asserts equality. The DF "match" is retained only as a labelled *comparison-workbook* check (not an ISACA benchmark).	Closed (was High)
B12	Design-factor formula	□□ `DF1–DF4` weighted sum + normalisation + target mapping (25/50/75) matches the *Design Guide* example logic (`[VER-P]` Souza Neto et al. quote of *Design Guide* §7.2.4.1), but the prototype's matrix constants and `BENCHMARK_ASSESSMENT_STATE` are **demo/interview-derived** (`src/data/benchmarkData.ts`), not ISACA data.	Medium

4.3 Recommended fix (cheap, high value)

> **Status 2026-10-13: this recommendation has been implemented.** The gate is configurable (`aggregate-85` / `all-l-or-f` / `iso-staged`) and displayed in the UI, all-N.A. levels are marked *Insufficient Evidence*, and the test banner is relabelled with a separate independent reference-calculation oracle added. The remaining action is the thesis write-up (name the construction honestly and pick the gate to defend).

Make the aggregation rule **configurable and visible**, and state it in the thesis as an explicit design decision:

0. **Name the construction honestly**: an "*adaptation of the COBIT 2019 capability-level scale using the ISO/IEC 33020:2019 N/P/L/F percentage bands*", never "the COBIT 2019 method". Justify the adaptation with ISACA's own words: COBIT is a reference model and "*it is the user's choice to determine how it is utilized*" (Elue 2020), and COBIT 2019 ships **no PAM and no normative algorithm** (§4.1a).
1. Offer **two gates**: **(i) all-activities-L-or-F** (matches the published proposal) and **(ii) aggregate >85%** (current). Report both in results for the same dataset — this *is* a research finding, not a bug.
2. For `N.A.`-only levels, mark the level **"incomplete/insufficient evidence"**, not N.
3. Rename the existing test file's banner from "GROUND TRUTH" to "reference/regression test"; add a **separate** documented reference-calculation fixture (see §8.3).

4.4 Corrections to existing drafts (Focus B)

Where	Claim in draft	Problem	Correct statement
`01-literat` `review.` §3.4	Capability names "Incomplete/Initial/Managed/Defined/Quantita Managed/Optimizing", labelled "*verified from COBIT 2019 I&M*"	**Wrong ladder mixed in.** The framework gives **six descriptive characteristics** (§4.1c). The *canonical* capability-level names are ISO/IEC 33020:2019 §5.2 — **0 Incomplete · 1 Performed · 2 Managed · 3 Established · 4 Predictable · 5 Innovating**. The "Initial/Managed/Defined/Quantitative/Optimi words belong to COBIT's **separate focus-area maturity** ladder (I&M §6.4.3, fig. 6.3)	Present the six capability characteris verbatim; if a name is used, use the ISO/IEC 33020:201 §5.2 names; keep the maturity

			names for focus areas only
01-literat review.: §3.5 / 03-decisio log.md	"level achieved only if **all activities receive F**"	Published proposal says **all L or F** — different	Correct to "all activities rated **Largely or Fully**" (Souza Neto et al. 2019)
01-literat review.: §3.5	Scale labelled "ISO/IEC 33000-based"	Attribution should be: **COBIT 2019 I&M §6.4.2**, whose percentage bands in turn mirror **ISO/IEC 33004:2015** and are identical to **ISO/IEC 33020:2019 §5.3** (for the precise boundaries)	Fix attribution; cite ISO/IEC 33020:2019 §5.3 for the exact bands
01-literat review.: §3.6 / 02-... 05-PRD	"Target-Level Cascade Algorithm (**Almeida 2019**)"	Authors are Souza Neto, **Almeida**, Mira da Silva; it is an ISACA Industry News **proposal**	Correct citation + label "proposal"
§3.6 (already good)	Scale called "asumsi operasional"	□ Already correct — keep this discipline; apply it to the "vs Excel" test claim too	—
01-lit-review.: §3.7	"JKU academic paper", "NETBankAudit (2025) 1,378 evaluations"	Unverifiable as written	[2ND-UNV] until primary located

4.5 Focus-B gap table

Element	Claim	Evidence	Status	Verdict
B-A	COBIT 2019 has **no** official PAM	ISACA Journal 2021	[VER-P]	**Solid**
B-B	Scale F/L/P/N + thresholds are official	ISACA COBIT-Focus 2019	[VER-P]	**Solid**
B-C	No official aggregation formula exists	ISACA 2021 + absence in I&M	[VER-P]	**Solid (this is the gap)**
B-D	Aggregation must be documented + evidence-anchored	ISACA 2021 (milestones/expected evidence)	[VER-P]	**Solid (supports Focus D)**
B-E	A per-level gate exists	Souza Neto et al. 2019	[VER-P]	**Solid (proposal)**
B-F	Prototype's aggregate gate == published gate	Engine code	[VER-P] (repo)	**False — different**
B-G	Prototype "matches Excel 100%"	Test file	[VER-P] (repo)	**False — unsupported**

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5. Focus C — Existing Tools and Software Comparison

> **Label discipline:** rows marked `[VER-P]` were fetched this session; `[2ND-UNV]` are inherited from `02-system-design-research.md` §2 and **must be independently re-verified** before publication. Absence of a feature = "not found in this source", **not** "does not exist."

5.1 ISACA official artefacts

Artefact	Type	What it does	Assessment capability?	Status
COBIT 2019 Tool Kit (free ZIP: overview, exec summary, FAQ, objectives/practices/activities `.xlsx`, RACI matrix)	`.zip` + PDFs + `.xlsx`	Reference material; the objectives/practices/activities file is a **flat data dump** (sheets `Objectives`, `Objectives-Practices`, `Activities`); the `Management-Awareness-Diagnostic.xls` is a self-diagnostic, not a 0–5 calculator	**No**	`[VER-P]` (ZIP downloaded, contents verified 2026-10-13)
COBIT 2019 Design Guide Tool Kit (a.k.a. Governance System Design Toolkit; paid)	`.xlsx` (27 sheets)	Weights the 40 objectives by **10 of the 11 design factors** (DF11 Enterprise Size deliberately excluded) via automatic `SUMPRODUCT`/`MROUND` formulas → prioritised design (initial/refined/concluded scopes); target capability levels are set **manually**; **single-assessment** (no history)	**No** — design/prioritisation only; rigid, protected formulas and expert-panel mapping values (criticised in Utomo et al. 2022; Amore et al. 2023)	`[VER-P]` (workbook downloaded, formulas/sheets verified)
COBIT 2019 Implementation Guide / I&M / G&M Objectives	Books	Describe the framework, CPM constructs, **1,202 activities / 231 practices / 40 objectives / 5 domains**	Guidance only, no computational instrument	`[VER-P]`
Introducing COBIT 2019: Major Differences with COBIT 5	PDF (in free kit)	States the PAM/ISO-33000 model was **replaced** by a CMMI-inspired model; no separate PAM publication is provided	n/a	`[VER-P]`
COBIT 5 PAM, Assessor Guide	Books	COBIT 5 process assessment (ISO/IEC 15504)	COBIT **5** only; superseded	`[VER-P]`

Central finding (upgraded): ISACA's only official *digital, computational* artefact is a **design/prioritisation** tool that (a) does **not** compute 0–5 capability/maturity, (b) is single-assessment (no history/versioning), and (c) uses protected formulas whose scores even ISACA says are directional. **There is no official assessment engine — verified at the artifact level this session.** `[VER-P]`

5.2 Commercial GRC / IT-assurance platforms (re-verified this session)

Platform	COBIT 2019 native?	Implements COBIT CPM	Evidence mgmt	Reporting / history	Key limitation	Status

		(0–5)?				
Hyperproof	COBIT template + crosswalking (180+ frameworks)	**No evidence** (control status)	Yes	Yes / partial	Control-status model	[VER-S]
ServiceNow GRC	via ISACA content pack (community thread cites **COBIT 5**)	**No evidence**	Yes	Yes / Yes	Version ambiguity	[VER-S]
Archer IRM	none found on official page	**No evidence**	Yes	Yes / Yes	Control/risk model	[VER-S]
MetricStream	third-party only	**No evidence**	unconfirmed	Yes / unconfirmed	—	[2ND-UNV]
ZenGRC (RiskOptics)	dedicated COBIT page	**No evidence**	Yes	Yes / unconfirmed	Version unconfirmed	[VER-S]
Diligent (HighBond)	none found	**No evidence**	Yes	Yes / Yes	Control/risk model	[VER-S]
SAP GRC (Pathlock)	none found	**No evidence**	Yes	Yes / unconfirmed	Page blocked — **not absence**	[2ND-UNV]
LogicGate / OneTrust	none found	**No evidence**	Yes	Yes / unconfirmed	Control/risk model	[VER-S]
Complyance	not in framework list	**No evidence**	Yes	Yes / unconfirmed	COBIT in blog only	[VER-S]
Drata / Secureframe	no native COBIT (Secureframe: only inside C5)	**No evidence**	Yes	Yes / unconfirmed	Compliance, not COBIT	[VER-S]
Vanta / Optro(AuditBoard) / Wolters Kluwer TeamMate	—	—	—	—	**Blocked/404 — could not verify, NOT absence**	[2ND-UNV]
CMMI Cybermaturity Platform (ISACA/CMMI)	COBIT not the model	**CMMI maturity** (0–5), not COBIT CPM	Yes	Yes / Yes	Scores CMMI cybermaturity	[VER-S]

Pattern (verified): commercial GRC tools use **control-status or risk scoring** (SATISFIED/PARTIAL/GAP, inherent/residual risk). **None verifiably implements COBIT 2019 CPM capability levels (0–5).** Where COBIT content exists it is a licensed pack or a crosswalk, usually for an unconfirmed version. [VER-S]

5.3 Dedicated COBIT tools & open source

Tool	Type	Capability calc?	Notes	Status
ITS Governance	Service+platform	Claimed, formula undocumented	—	[2ND-UNV]
Complyance	SaaS	Not confirmed	Monitoring model	[2ND-UNV]
Dynaflow	Commercial	Not confirmed	Control library (>1,200 controls)	[2ND-UNV]
ControlWeave	Open source (AGPL-3.0)	Not confirmed (NIST/ISO model)	Evidence versioning (SHA-256) — closest analogue	[2ND-UNV]

A-COBIT-2019-Risk-Assessment	Open source (GPL-3.0)	No (risk assessment)	—	[2ND-UNV]
evaluador-ti	Open source	Maturity scoring (COBIT/ITIL)	Unmaintained	[2ND-UNV]
CMMI Cybermaturity Platform	ISACA/CMMI commercial	Yes — but **COBIT 5**, cyber-scoped	—	[2ND-UNV]

5.4 Academic prototypes (re-verified at repository/DOI level)

Study	Year	Venue/Inst.	COBIT	Dev method	Scoring	Evaluation	Evidence mgmt	Status
Mandianga S. S.	2023	S1 thesis, UPN Veteran Jakarta (repository)	2019	Agile Scrum (4 sprints), PHP/Bootstrap/M	**Skala Guttma + NPLF** status	Black-box UAT (all pages pass)	No	[VER-S] — **closest prior art**
Abdullah, Putra & Kartika	2023	J. Sistem Informasi & Ilmu Komputer, DOI 10.59581/jusiik-widyakarya.v1i4.1319	2019	SCRUM, Laravel	not exposed in abstract	UAT	No	[VER-S]
Anwar & Harits	2025	J. Informatika Utama 3(1), DOI 10.55903/jitu.v3i1.271	2019	RAD, PHP/Java/MySQL	Capabili score + graph	not in abstract	No	[VER-S]
Noor, M. S.	2021	Thesis, ITTP (repository)	2019 (**BAI only**)	Prototyping	capabilit assessm	Black-box	No	[VER-S]
Pamuji, Putra & Wahdiniwa	2022	J. TK3TI, DOI 10.34010/jtk3ti.v8i2.8131	2019 + ISO/IE 15504-2	assessment model	capabilit level	audit application	No	[VER-S]
Kurniawan et al.	2025	INTI Nusa Mandiri, DOI 10.33480/inti.v19i2.6389	**COBIT 5** (EDM)	integrated audit system	**still manual**	20 respondent	No	[VER-S]
Indrawati, Rozas & Wahyudi	2023	Edu Komputika, DOI 10.15294/edukomputika.v10i1.597	2019	R&D instrument	maturity instrume	R&D validation	No	[VER-S]
~~Cobvapps (UB)~~	—	—	—	—	—	—	—	**[NF] — not found; likely misattril Remove from drafts.**

Consequence for novelty (important): web-based COBIT 2019 assessment tools **already exist** (Mandiangan 2023; Abdullah et al. 2023; Anwar & Harits 2025; Noor 2021). The thesis therefore **cannot** claim novelty in *kind*. The defensible novelty must be in **one of**:

- the **scoring model** (Mandiangan itself lists its limited parameters and suggests a Likert scale as future work → an explicit, documented CPM aggregation rule is a real improvement);

- **process coverage / design-factor-driven prioritisation** (Noor = single domain; Kurniawan = COBIT 5);
- **evidence traceability** (absent in all surveyed prototypes);
- **evaluation rigour** (prior art = black-box/UAT only; add reference-calculation + expert review + SUS);
- **domain context**.

5.5 What exists vs. what the thesis contributes

Capability	Commercial GRC	ISACA official	Dedicated COBIT	Open source	Academic	This thesis
COBIT 2019 content	□	□ (design)	□	partial	□	□ (subset)
Official capability **instrument**	□	□	□	□	□	□ (not official — must say so)
Automated CPM calculation	□	□	unverified	□	partial	□
Transparent/inspectable scoring	□	n/a	unverified	□	□	□
Evidence-to-rating traceability	partial	□	unverified	partial	□	□
Assessment **management** (history/comparison/report)	partial	□	unverified	partial	□	□ (design goal)

5.6 Focus-C gap table

Element	Claim	Evidence	Status
C1	Official tool = design only	ISACA 2020	`[VER-P]`
C2	Commercial GRC ≠ capability assessment	Draft matrix	`[2ND-UNV]`
C3	No open-source COBIT 2019 CPM engine found	Draft matrix	`[2ND-UNV]`
C4	Academic prototypes lack transparent/documented scoring	Mandiangan uses Guttman+NPLF (documented *in the thesis*, not in the tool); Capability calc is not plainly inspectable in the others	`[VER-S]`
C5	No prototype links scoring to **evidence** or manages **multi-assessment**	Verified across #1–#8	`[VER-S]` — the pivotal row
C6	Web-based COBIT 2019 assessment tools are novel	**False** — several exist	`[VER-S]` — **do not claim**

> **Rule for the thesis:** never write "no tool exists." Write "*no tool with [feature] was found in the sources surveyed on 2026-10-13 (see Table X).*"
>
> **Rule for novelty:** do not claim "first web-based COBIT 2019 assessment system." Claim the **specific** differentiator (transparent CPM aggregation + evidence traceability + assessment management + stronger evaluation).

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6. Focus D — Assessment Management as an IS Concern

6.1 Why this is an Information Systems problem (not a feature list)

An assessment is not a one-off calculation; it is an **information artefact with a lifecycle**. The IS concerns are:

IS concern	Question the system must answer
Identity & scope	Which organisation, which period, which objectives, which target levels?
Multi-assessment	Can the same organisation be assessed repeatedly (year/unit/department)?
History	What was rated before, by whom, on what evidence, when?
Comparison	Is capability improving or regressing between periods/units?
Traceability	For each rating, *which* evidence supports it, and can a third party reproduce it?
Reporting	Can the result be communicated to non-experts (management), reproducibly?

This is exactly the "data → information → audit conclusion" chain that audit standards govern: the IIA *Global Internal Audit Standards* (2024) organise Domain V (*Performing Internal Audit Services*; Principles 13/14/15 = Plan Engagements Effectively / Conduct Engagement Work / Communicate Engagement Results and Monitor Action Plans), and **Standard 14.6 "Engagement Documentation"** (verified from the official IIA text) governs documenting the engagement. ISACA's **ITAF Standard 1205 "Evidence"** (4th ed., 2020) requires that "*IT audit and assurance practitioners shall obtain sufficient and appropriate evidence to draw reasonable conclusions.*" `[VER-P]` (both verified this session; cite ITAF/IIA by edition).

ISACA itself names the missing layer: the 2021 Journal article says the maturity model must culminate in "*a tool that allows an assessment of the organisation and creates the proper reporting in a language that top-level management can understand.*" `[VER-P]`

6.2 Requirement classes (IS framing)

Class	Requirement	Anchor
Assessment identity	assessment ID, org, unit, period, assessor, target levels	CPM step 2–3
Rating capture	per activity (and per level), with comment + reason for N.A.	CPM step 5
Traceability	each rating ↔ ≥0 evidence items; evidence has source/type/date/status	CPM step 4–5 (`[VER-P]` ISACA: "traceability must be maintained")
Scoring transparency	the method and its parameters are visible in the UI and the export	Gap under B (§4.2)
Multi-assessment & history	compare assessments over time/units	IS concern (§6.1)
Reporting	export (PDF/print) with method, assumptions, gaps	ISACA step 6
Reproducibility	re-running the same data yields the same result, with the method recorded	IIA 2024 documentation principle

6.3 Prototype honesty check

- **Updated 2026-10-13:** the client-side SPA now has **minimal localStorage persistence** (assessment registry + compact snapshots) and a **baseline→candidate comparison page**. So "multi-assessment / history / comparison" is now an **implemented capability**, not only a design goal.
- Benchmark dataset: **synthetic / interview-derived** — `src/data/benchmarkData.ts`, and the UI now labels it "*Data Demo*" rather than "Benchmark PANRB".
- Still to do: (a) finalise the seeded COBIT subset and its licensing note (Appendix B #6); (b) in the thesis, scope the persistence as *single-browser, client-side* (no server, no multi-tenant) and evaluate it as such. Do not claim more than is built.

6.4 Focus-D gap table

Element	Claim	Evidence	Status	
D1	CPM requires evidence traceability	ISACA 2019	`[VER-P]`	
D2	Evidence must be definable per activity	ISACA Journal 2021	`[VER-P]`	
D3	Audit documentation must be reproducible	IIA 2024 (principle); exact number `[NF]`	`[VER-P]`/`[NF]`	
D4	Tool should produce management-language reporting	ISACA Journal 2021	`[VER-P]`	
D5	Existing tools lack the management layer	Draft matrix	`[2ND-UNV]`	
D6	Prototype implements multi-assessment/history	Repo	`[VER-P]` (repo)	✅ **Yes (2026-10-13)** — registry, snapshots, and comparison page implemented

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7. Focus E — AI / RAG: Include, Simple, or Omit?

7.1 Recommendation

> **OPTIONAL-SIMPLE (fallback: OMIT).** A single direct LLM call that drafts a recommendation from the computed gap, always labelled "*AI-generated draft — assessor review required*". **No RAG pipeline. No retrieval metrics. No AI in the title, RQs, or method chapter.** `[OWN]`

Decision rule to state in the thesis: *AI is included only if it adds demonstrable value at low methodological cost; it must not be on the critical path of any research question.*

7.2 Why the RAG framing is now a liability

The existing `01-literature-review.md` is heavily RAG-weighted. Three problems surfaced when sampling its claims:

#	Claim in draft	Finding	Action
E1	"Eulerich et al. (2023), *Contemporary Accounting Research* (n=internal audit functions)" with barrier stats 45.9% / 34.1% / 46.5%	The CAR 2023 Eulerich et al. paper located is "*The Impact of Audit Technology on Audit Task Outcomes: Evidence for Technology-Based Audit Techniques*", **40(2), 981–1012, DOI 10.1111/1911-3846.12847** — it does **not** match the described adoption-barrier study	`[2ND-UNV]` — locate the true source or delete the stats
E2	"KPMG 2025 SOX Survey: controls in scope more than doubled in 2 years; 16 hours per control"	The KPMG survey exists, but the reported figure is **in-scope *systems* 17 → 40** (≈2.3×) and avg program cost **$2.3M**; automated controls **21% (FY22) → 17% (FY24)**. "16 hours/control" not found	Correct "controls" → "systems"; drop the 16-hr figure unless sourced
E3	RAG claims: "93.6% validate on isolated academic data" (Karakurt 2026); "NETBankAudit 2025: 1,378	Karakurt & Akbulut (2026) **is real** (Appl. Sci. 16(1), 368, DOI 10.3390/app16010368) but the 93.6% figure was not confirmed;	`[2ND-UNV]` both

evaluations"	NETBankAudit not found

The pattern ([OWN]): the RAG layer imported a **large, partly unverifiable** body of claims that are *not needed* to defend the assessment gap. Under examiner scrutiny this is the most fragile part of the draft — while the COBIT/CPM/evidence argument is the strongest.

7.3 If AI is kept, the minimal safe design

```
gap result (objective, current vs target, evidence count)
   → templated prompt + COBIT practice context (short, cited)
     → single LLM API call
       → draft text in an editable field
         → label: "AI-generated draft — assessor review required"
```

No vector store, no embeddings, no retrieval evaluation. Value is demonstrable in a demo; risk is contained.

7.4 Focus-E gap table

Element	Claim	Evidence	Status	
E1	RAG is required for COBIT assessment	No source requires it	[OWN]	**Rejected**
E2	AI improves recommendation quality	Not established in draft	[2ND-UNV]	Unproven
E3	AI adds demo value at low cost	Reasonable	[OWN]	Keep as optional
E4	RAG-literature claims are solid	3 source problems	[2ND-UNV]	**Trim**

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8. Focus F — DSRM Methodology and Evaluation Plan

8.1 Methodology (verified references)

Peffers, Tuunanen, Rothenberger & Chatterjee (2007), *A Design Science Research Methodology for Information Systems Research*, JMIS 24(3), 45–77, DOI **10.2753/MIS0742-1222240302**. [VER-P]

Six activities: (1) problem identification & motivation; (2) definition of solution objectives; (3) design & development; (4) demonstration; (5) evaluation; (6) communication. Four entry points: problem-centred; objective-centred; design/development-centred; client/context-initiated. [VER-P]

Supporting: **Hevner et al. (2004)**, MISQ 28(1), **75–105**, DOI **10.2307/25148625** — the IS-research framework + **seven guidelines** + **five evaluation-method categories** (Table 2): Observational, Analytical, Experimental, **Testing** → **Functional (Black-Box)**, Descriptive. [VER-P] *(Note: the "three-cycle view" is **not** from 2004 — it is Hevner (2007), SJIS 19(2), Art. 4. Do not mis-attribute it.)*

Contribution framing — Gregor & Hevner (2013), MISQ 37(2), 337–355, DOI **10.25300/MISQ/2013/37.2.01** [VER-P]. 2×2 knowledge-contribution framework: Invention / **Improvement** / **Exaptation** / Routine Design. Because web COBIT 2019 assessment tools already exist (§5.4), this thesis should frame its contribution as **Improvement** (a better artefact for a known problem) or **Exaptation** (a known method applied in a new context) — **not** Invention. [OWN]

8.2 Evaluation framework

Venable, Pries-Heje & Baskerville (2016), *FEDS*, EJIS 25(1), 77–89, DOI **10.1057/ejis.2014.36**. [VER-P] Dimensions: formative/summative × artificial/naturalistic. Strategies: Quick&Simple; Human Risk & Effectiveness; Technical Risk & Efficacy; Purely Technical Artefact.

Recommended hybrid for this thesis: *Technical Risk & Efficacy* (artificial — reference-calculation tests of the scoring engine) **+** *Human Risk & Effectiveness* (naturalistic — expert review and SUS). [OWN]

8.3 Evaluation plan (what each method proves — and does not)

Method	What it proves	What it does not prove	Reference
Black-box functional testing	Observable outputs conform to expected outputs for chosen inputs; defects discovered	Absence of defects; correctness of untested paths	Hevner et al. 2004 Table 2 ([VER-P]); ISO/IEC/IEEE 29119-4:2021, DOI 10.1109/IEEESTD.2021.9591574
Reference-calculation validation (test oracle)	The engine reproduces an independently derived expected result	That the *method itself* is valid	Barr et al. 2015, DOI 10.1109/TSE.2014.2372785 ([VER-P])
Expert review / content validity	Domain experts judge the COBIT mapping and scoring model as appropriate	Statistical accuracy	Lynn 1986 (I-CVI), DOI 10.1097/00006199-198611000-00017; Polit & Beck 2006, DOI 10.1002/nur.20147 ([VER-P])
SUS	*Perceived* usability (ordinal comparison)	Correctness or functionality	Brooke 1996, DOI 10.1201/9781498710411-35; Lewis 2018, DOI 10.1080/10447318.2018.1455307; Fiandhika & Kelana 2024, DOI 10.1051/e3sconf/202448303008 ([VER-P])

SUS practicals: 10 items; positive items (1,3,5,7,9) score = (position − 1); negative items (2,4,6,8,10) = (5 − position); sum × 2.5 → 0–100. Acceptability (Bangor et al. 2008): <50 not acceptable, 50–70 marginal, >70 acceptable; benchmark ≈ 71. Adjective scale (Bangor et al. 2009). Missing item → midpoint (3). [VER-P] for items/scoring; [VER-S] for bands.

Sample size & reporting (the part students get wrong): there is **no universal minimum** for SUS, but the *precision of the mean* is poor at small N. Recommend **N ≥ 20–30** respondents and always report a **confidence interval** (SUS SD ≈ 17.7–21; at n=20 the 90% margin ≈ ±8 points). Label the study *summative, naturalistic* per FEDS. [VER-S]

Expert review — scope caution: Aiken's V (Aiken 1985, DOI 10.1177/0013164485451012) and the CVI family (Lawshe 1975, DOI 10.1111/j.1744-6570.1975.tb01393.x; Polit & Beck 2006, DOI 10.1002/nur.20147) validate the **questionnaire/instrument**, **not the software**. Do not write "the system has a content-validity index of 0.89" — that is a category error. Validate the instrument with experts; evaluate the system by testing / reference comparison / usability. [VER-P] + [OWN]

Inter-rater agreement (if two assessors score the same cases): Cohen's κ (Cohen 1960, DOI 10.1177/001316446002000104) with Landis & Koch (1977, DOI 10.2307/2529310) bands (<0 poor · 0–0.20 slight · 0.21–0.40 fair · 0.41–0.60 moderate · 0.61–0.80 substantial · 0.81–1.00 almost perfect). Use **weighted kappa / ICC** for the ordinal 0–5 levels; use **Fleiss' κ** for >2 raters. [VER-P]

8.4 The credibility problem in "comparison with manual / vs Excel"

`02-system-design-research.md` §EV and `05-PRD` line 471 claim the test script verifies "akurasi matematis vs Excel". **There is no ISACA assessment workbook** to compare against (the official artefact is a *design* tool). Therefore the "reference" must be one of:

1. **Recommended:** an **independently derived reference calculation** — compute expected level results by hand/independent script from the *documented CPM rules* (the F/L/P/N bands + the chosen gate), store inputs+outputs as a fixture, and assert equality (a golden/reference test). Document the derivation.
2. **Acceptable, if labelled:** the **PANRB course workbook** as "*a comparison workbook, not an official ISACA benchmark*" — which the PRD already says (lines 43, 429).
3. **Not acceptable:** calling any of these "matching the ISACA Excel 100%."

Precedent exists and should be cited. System-vs-manual reference comparison is an established validation pattern in Indonesian IS research — e.g., a 2026 palm-oil reporting app compared system output against manual Excel over 177 data points (paired test + regression); a credit-scoring SPK reported 97.78% agreement with manual calculation. `[VER-S]` For ordinal capability levels, prefer **Wilcoxon signed-rank** or **ICC** over a paired *t*-test.

Honesty requirement: if the engine re-implements the same formula you also applied by hand, the comparison proves **implementation equivalence (verification)**, not that the COBIT result is *correct* in a business sense. Say exactly that. `[OWN]` This is the difference between a defensible S1 evaluation and an overclaim.

8.5 Focus-F gap table

Element	Claim	Evidence	Status
F1	DSRM is the right method for a design artefact	Peffers 2007	`[VER-P]`
F2	Evaluation must be designed, not ad hoc	FEDS	`[VER-P]`
F3	Black-box + SUS + expert review is a valid bundle	Hevner 2004; precedents	`[VER-P]`
F4	SUS measures usability, not correctness	Lewis 2018; Fiandhika 2024	`[VER-P]`
F5	Reference-calculation must be an independent oracle	Barr 2015	`[VER-P]`
F6	Current "vs Excel" claim is credible	Repo	`[VER-P]` **No — must be reframed**

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9. Gap Tables, Corrections Register, and Bab 1 Scaffold

9.1 Consolidated gap table (the defensible gap)

ID	Gap element	Evidence anchor	Status	Role in thesis
G1	No official COBIT 2019 PAM / assessment instrument	ISACA Journal 2021 (`[VER-P]`)	Solid	Problem statement
G2	Official digital artefact is design-only	ISACA Industry News 2020 (`[VER-P]`)	Solid	Problem statement

G3	CPM aggregation is **undefined by ISACA**	ISACA Journal 2021 (`[VER-P]`)	Solid	**Design contribution**
G4	CPM ratings require **evidence traceability**	ISACA 2019 (`[VER-P]`)	Solid	**Design contribution**
G5	Assessment is a lifecycle artefact (multi/history/compare/report)	IS concern; ISACA reporting (`[OWN]`/`[VER-P]`)	Solid	**IS contribution**
G6	Commercial GRC lacks capability computation	Draft matrix (`[2ND-UNV]`)	Needs re-verify	Existing-system gap
G7	Academic prototypes lack transparent scoring + evidence	Draft matrix (`[2ND-UNV]`)	Needs re-verify	Existing-system gap
G8	"Excel → web" per se	—	Rejected	**Not a gap**

One-sentence gap statement (for the thesis):

> *COBIT 2019 provides a rating scale and a capability-level construct but **no official assessment instrument, no mandated aggregation formula, and no evidence-traceable assessment-management tool**; existing GRC products map controls and academic prototypes move the spreadsheet online without documented scoring or evidence linkage. This thesis designs and evaluates a web-based system that makes the CPM scoring explicit and inspectable and links every rating to audit evidence.*

9.2 Corrections register (claims to fix before submission)

#	Location	Issue	Action	Priority
1	`01-lit-review.md` §3.4	Capability names mis-stated as framework-verified	Replace with the six characteristics; label one-word names `[2ND-UNV]`	High
2	`01-lit-review.md` §3.5	"all activities receive F"	Correct to "all **L or F**"	High
3	`01-lit-review.md` §3.5	"ISO/IEC 33000-based"	Attribute to COBIT 2019 I&M (from ISO/IEC 33004:2015)	Medium
4	`01-lit-review.md` §3.6	"Almeida 2019"	→ Souza Neto, Almeida & Mira da Silva (2019), ISACA Industry News (proposal)	High
5	`02-...` §EV / `05-PRD` 471 / `engine.test.ts`	"100% match vs Excel"	Reframe as reference-calculation/regression test; delete "Excel match"	**Critical**
6	`01-lit-review.md` line 10	Eulerich 2023 CAR stats	Verify or delete	High
7	`01-lit-review.md` line 40	KPMG "controls doubled / 16 hrs"	Correct to systems 17→40; drop 16-hr	Medium
8	`01-lit-review.md` §3.7	JKU / NETBankAudit / 93.6%	`[2ND-UNV]`; verify or delete	Medium
9	`landing/benchmarkData.ts` framing	Synthetic data implied as benchmark	Label "synthetic demo data"	High
10	`latar-belakang.md` ¶1	Indonesian regulation as direct justification	Keep as **context**, not justification for the COBIT tool	High
11	Whole draft	RAG centrality	Demote to optional future work	High
12	Title/RQ	"Capability Level" vs "Maturity Level"	Keep "Capability Level" (per process) — correct	—
	`02-system-design-`		**Not found; likely misattributed** —	

13	`research.md` §2.5	"Cobvapps (UB)" row	remove unless a primary record is produced	High
14	Draft novelty claims	Implied "first web COBIT 2019 tool"	Web COBIT 2019 tools already exist (Mandiangan 2023; Abdullah 2023; Anwar & Harits 2025; Noor 2021) — claim **Improvement/Exaptation** only	**Critical**
15	DSRM chapter	If it cites "Hevner et al. 2004" for the three-cycle view or page range 75–106	Correct to Hevner (2007), SJIS 19(2); pages **75–105**	Medium
16	Evaluation plan	If it claims "expert validation proves the software works"	Experts validate the **instrument**, not the software	High
17	Evaluation plan	SUS sample size / reporting	Use N ≥ 20–30 and report a confidence interval	Medium
18	`02-...` §2.5 / `03-decision-log` §3.4	Mandiangan characterised as "cuma mindahin Excel ke web, rumus tidak transparan"	It uses a **documented** Guttman+NPLF method (in the thesis); soften to "scoring not inspectable *within the tool* / limited parameters (author-acknowledged)"	High
19	`04-regulasi-...md` line 12	"*Tidak ada satu pun regulasi Indonesia yang menyebut 'COBIT*'"	**FALSE as written.** COBIT is *named* (as an optional best practice, not a mandate) in POJK 38/2016 (Penjelasan I. umum), SEOJK 21/2017 (Lampiran §5.2, §7.6.1), PermenPANRB 5/2018 (revoked), and PermenPANRB 59/2020 (Lampiran). Correct to: **no** Indonesian regulation **mandates** COBIT; it appears only as an optional best-practice reference, chiefly in revoked instruments	High
20	`04-...` §1.2 / this report's earlier note	Perpres 132/2022 treated as an amendment to Perpres 95/2018	**Not an amendment.** Perpres 132/2022 = *Arsitektur SPBE Nasional* — a separate Perpres issued under Pasal 9(4)/74 of Perpres 95/2018	Medium
21	`04-...` §1.4	PermenPANRB 59/2020 cited as the current SPBE-evaluation instrument	**Revoked** by **PermenPANRB No. 8/2026** (*Evaluasi Kinerja Pemerintah Digital*; current index = *Indeks Pemerintah Digital*, ~20 indicators). Cite 8/2026 for current law; 59/2020 (47 indicators, 1–5 maturity) historically only	**High (currency)**
22	`04-...` §1.3	Permenkominfo 16/2022 *Audit TIK* definition quoted with "standar dan praktik-praktik terbaik"	That phrase is the **Lampiran's definition of "kriteria"**, not the Pasal 1 definition of *Audit TIK* ("kriteria dan/atau standar yang telah ditetapkan"). Separate the two quotes	Medium

23 `04-...` §1.6.3	"SEOJK 21/2017 cites ISO & SNI and **does not** mention COBIT"	**Wrong at document level** — only §1.7(b) says ISO/SNI; §5.2 and §7.6.1 **do** name COBIT/ITIL/IEC	Medium
24 `04-...` §2.2.2	Sebatik reference dated 2025	It is **Akbar & Saputra (2023), *Sebatik* 27(2): 589–605**, DOI 10.46984/sebatik.v27i2.2336	Low
25 `01-.../02-.../`dashboard	"Design Guide Tool Kit" described as an assessment tool or the reference for the "vs Excel" claim	The Design Toolkit is a **design/prioritisation** workbook (10/11 design factors, protected formulas, no 0–5 computation, single-assessment) — it cannot be the oracle for capability results	**Critical**

9.3 Recommended `latar-belakang.md` structure (Bab 1)

¶	Move	Point to make	Cite
1	IT governance need	IT value/risk/resource management require governance	ISACA 2019
2	COBIT 2019 as reference	40 objectives, 5 domains, design factors, CPM (0–5), F/L/P/N	ISACA 2019/2020
3	**The instrument gap**	No PAM for COBIT 2019; official toolkit is design-only; **no mandated aggregation**	ISACA Journal 2021; ISACA Industry News 2020
4	**The evidence problem**	Ratings must be evidence-based and traceable; evidence is scattered; third parties must be able to reproduce	ISACA 2019 (traceability); IIA 2024 (documentation)
5	**The management problem**	Assessment is repeated over time/units; results must be compared and reported to management	ISACA Journal 2021
6	Existing tools	GRC maps controls; prototypes move Excel without transparency/evidence	§5 of this report
7	Research response	DSRM + explicit scoring + evidence traceability + evaluation	Peffers 2007; FEDS
8	*Optional* Indonesian context	PP 71/2019 Pasal 22 audit-trail **obligation** + Permenkominfo 16/2022 Audit-TIK **definition** as *context*, **not** justification. Never claim regulation mandates COBIT; cite current law (**PermenPANRB 8/2026**, not 59/2020)	`[VER-P]` PP 71/2019 Pasal 22(1); Permenkominfo 16/2022 Pasal 1(2); §9.5

Do not write: "SPBE evaluation requires COBIT 2019." SPBE evaluation uses PermenPANRB indicators; COBIT is a voluntary reference. (`[OWN]` — consistent with `04-regulasi-indonesia-cobit.md` claims, which still need verification of current status.)

9.4 Title & RQ recommendations

- **Keep** the title (it names an artefact + a differentiator).
- Consider tightening the third RQ from "usability" to include **correctness/traceability validation**, e.g.: *"Bagaimana validitas perhitungan CPM dan keterlacakan evidensi sistem dinilai melalui pengujian black-box, validasi terhadap perhitungan referensi independen, expert review, dan SUS?*"
- **Remove** any RQ that implies AI/RAG.

9.5 Indonesian regulation — verified facts and the correct use (access 2026-10-13)

Verified against JDIH BPK / peraturan.go.id official PDFs, including full-text keyword checks ("COBIT" occurrences). This section exists so the thesis neither over-claims nor under-claims the regulatory angle.

#	Instrument	Status	Names "COBIT"?	Correct use in the thesis
R1	**PP 71/2019** (PSTE)	In force	**No**	Pasal 22(1) verbatim: "*Penyelenggara Sistem Elektronik* ***wajib menyediakan rekam jejak audit*** *terhadap seluruh kegiatan penyelenggaraan Sistem Elektronik*." → motivation for **evidence traceability**, not for COBIT
R2	**Perpres 95/2018** (SPBE)	In force	**No** (0 occurrences)	Pasal 1(2) defines Tata Kelola SPBE as "pengaturan, pengarahan, dan pengendalian" (structural analogue of COBIT's EDM); Pasal 3(c) puts **Audit TIK** in scope; Pasal 55(5) delegates Audit-TIK policy to a ministerial regulation
R3	**Perpres 132/2022**	In force	**No**	*Arsitektur SPBE Nasional* — a **separate** Perpres (legal basis Pasal 9(4)/74 of Perpres 95/2018), **not** an amendment to it
R4	**Permenkominfo 16/2022**	In force	**No** ("standar dan praktik-praktik terbaik" only, in the Lampiran's definition of *kriteria*)	Pasal 1(2) defines *Audit TIK* as "*proses yang sistematis untuk memperoleh dan mengevaluasi* ***bukti secara objektif*** *… menetapkan* ***tingkat kesesuaian*** *… dengan kriteria dan/atau standar*" → motivation for **evidence-based capability measurement** against criteria
R5	**PermenPANRB 59/2020**	**Revoked** by **PermenPANRB 8/2026**	**Yes** (Lampiran, one maturity model among several)	SPBE evaluation = 4 domains / 8 aspects / **47 indicators** / maturity 1–5; **not** COBIT. Cite only historically
R6	**PermenPANRB 8/2026** (*Evaluasi Kinerja Pemerintah Digital*)	In force; revoked R5	**No**	**Current** instrument (~20 indicators); "Indeks SPBE" renamed "Indeks Pemerintah Digital"
R7	**Pedoman MenPANRB**	In force	**No**	Technical guide for SPBE evaluation; still the 47-

R7	**3/2024**	In force	**No**	indicator instrument per 59/2020
R8	**POJK 38/2016** · **SEOJK 21/2017** · **PermenPANRB 5/2018**	**Revoked**	**Yes** (as an example, non-mandatory)	Evidence that COBIT is *recognised* in Indonesia — but only as one best-practice option, in revoked instruments
R9	**POJK 11/POJK.03/2022**	In force	**No**	Pasal 2(2) lists 7 factors that **resemble** COBIT Design Factors but are **not labelled** COBIT; Pasal 54(4) requires annual internal IT audit

Correct statements (write these):

- "No Indonesian regulation **mandates** COBIT (any version); where COBIT appears it is one optional best practice among ISO/SNI/ITIL, mostly in revoked instruments." `[VER-P]`
- "SPBE evaluation uses **PermenPANRB** indicators and its own 1–5 maturity scale — **not** COBIT processes; COBIT is a voluntary, deeper self-assessment complement." `[VER-P]`
- "PP 71/2019 Pasal 22 and Permenkominfo 16/2022 establish the **obligations** (audit trail; evaluate objective evidence against criteria) but provide **no measurement instrument** — that instrument gap is the thesis's *context*, not its justification." `[VER-P]`

Wrong statements (do not write):

- ☐ "No Indonesian regulation mentions COBIT." *(false — see R5, R8)*
- ☐ "Perpres 132/2022 amended Perpres 95/2018." *(false — see R3)*
- ☐ "SPBE evaluation uses COBIT 2019." *(false — see R5/R6)*
- ☐ "COBIT 2019 is required because of [regulation]." *(no such requirement exists)*
- ☐ "PermenPANRB 59/2020 is the current SPBE instrument." *(revoked by PermenPANRB 8/2026)*
- ☐ "The PERMENPANRB course workbook defines the N/P/L/F method." *(`[NF]` — no such official workbook was found; the method's origin is ISO/IEC 15504→33020 / COBIT PAM)*

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10. Final Source List

Access date **2026-10-13** for all URLs. Labels: `[VER-P]` primary verified this session; `[VER-S]` secondary verified; `[2ND-UNV]` unverified; `[NF]` not found.

10.1 ISACA — official / published

1. ISACA. *COBIT 2019 Framework: Introduction and Methodology* (ISBN 978-1-60420-763-7; §6.3, §6.4.1–6.4.3, pp. 38–39: CPM, CMMI alignment, "no PAM", rating scale, focus-area maturity); *Governance and Management Objectives* (ISBN 978-1-60420-764-4; §3.4, p. 20: "*a process reaches a certain capability level as soon as all activities of that level are performed successfully*"); *Design Guide* (ISBN 978-1-60420-765-1; Ch. 6: the toolkit); *Implementation Guide* (ISBN 978-1-60420-766-8). 2018/2019. https://www.isaca.org/resources/cobit — `[VER-P]` (specific clauses; text read from publicly available PDF copies of the books — cite via the ISACA store/ISBN, not the mirror)
2. Dhulipalla, L. R. S. (2019, 16 Sep). *Using COBIT 2019 Performance Management Model to Assess Governance and Management Objectives*. ISACA COBIT Focus. https://www.isaca.org/resources/news-and-

trends/newsletters/cobit-focus/2019/using-cobit-2019-performance-management-model-to-assess-governance-and-management-objectives (archive: web.archive.org/web/20200804121432/…) — [VER-P]
3. Elue, E. (2020). *Effective Capability and Maturity Assessment Using COBIT 2019*. ISACA Industry News. https://www.isaca.org/resources/news-and-trends/industry-news/2020/effective-capability-and-maturity-assessment-using-cobit-2019 — [VER-P]
4. Souza Neto, J., Almeida, R., & Mira da Silva, M. (2019, 18 Mar). *Defining Target Capability Levels in COBIT 2019: A Proposal for Refinement*. ISACA Industry News. https://www.isaca.org/resources/news-and-trends/industry-news/2019/defining-target-capability-levels-in-cobit-2019-a-proposal-for-refinement — [VER-P]
5. Harisaiprasad, K. (2020, 27 Apr). *COBIT 2019 and COBIT 5 Comparison*. ISACA Industry News. https://www.isaca.org/resources/news-and-trends/industry-news/2020/cobit-2019-and-cobit-5-comparison — [VER-P]
6. Gorgona, L. (2021, 28 Dec). *Building a Maturity Model for COBIT 2019 Based on CMMI*. ISACA Journal, 6. https://www.isaca.org/resources/isaca-journal/issues/2021/volume-6/building-a-maturity-model-for-cobit-2019-based-on-cmmi — [VER-P]
7. ISACA. (2019). *COBIT Design Factors*. Industry News. https://www.isaca.org/resources/news-and-trends/industry-news/2019/cobit-design-factors — [VER-P] (prior session)
8. ISACA. (2020). *COBIT Tool Kit Enhancements*. Industry News. https://www.isaca.org/resources/news-and-trends/industry-news/2020/cobit-tool-kit-enhancements — [VER-P] (prior session)
9. ISACA. (2022). *Getting Creative with Maturity Models and COBIT 2019*. ISACA Now Blog. https://www.isaca.org/resources/news-and-trends/isaca-now-blog/2022/getting-creative-with-maturity-models-and-cobit-2019 — [VER-P] (prior session)

10.2 ISO/IEC — standards

10. ISO/IEC 33020:2015 / 33020:2019 — *Process measurement framework for assessment of process capability*. https://www.iso.org/standard/78526.html — §5.2 level names (0 Incomplete · 1 Performed · 2 Managed · 3 Established · 4 Predictable · 5 Innovating) and §5.3 rating bands (N 0–≤15%; P >15–≤50%; L >50–≤85%; F >85–≤100%) [VER-P] (preview); §5.6 staged rule (lower levels Fully, level claimed Largely) via the 2015 model reproduced in Automotive SPICE PAM 3.1, Table 16 [VER-S]
11. ISO/IEC 33004:2015 — *Requirements for process reference, process assessment and maturity models*. https://www.iso.org/standard/54178.html — [VER-S]
12. ISO/IEC/IEEE 29119-1:2022; 29119-2:2021; 29119-3:2021; 29119-4:2021 (DOI 10.1109/IEEESTD.2021.9591574) — [VER-P]

10.3 Audit standards & evidence

13. IIA. (2024). *Global Internal Audit Standards* (Domain V — Performing Internal Audit Services; **Standard 14.6 "Engagement Documentation"**; Principles 13/14/15). https://www.theiia.org/en/standards/2024-standards/ — [VER-P] (exact numbering verified from official text, 2026-10-13)
14. ISACA. *ITAF™: A Professional Practices Framework for IT Audit*, **4th edition (2020)** — **Standard 1205 "Evidence"** (1205.1: "…shall obtain sufficient and appropriate evidence to draw reasonable conclusions."); 1200-series = 1201 Risk Assessment in Planning · 1202 Audit Scheduling · 1203 Engagement Planning · 1204 Performance and Supervision · **1205 Evidence** · 1206 Using the Work of Other Experts · 1207 Irregularities and Illegal Acts. — [VER-P] (verified from the official ITAF 4th-edition document, 2026-10-13). *Note: a 5th edition exists (2026); state the edition when citing.*
15. ISACA. (2013). *COBIT Assessor Guide: Using COBIT 5*. — cited by ISACA 2019 article — [VER-S]

10.4 DSRM & design-science evaluation

16. Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). JMIS 24(3), 45–77. DOI 10.2753/MIS0742-1222240302 — [VER-P]

17. Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). MISQ 28(1), 75–105. DOI 10.2307/25148625 — [VER-P]
18. Gregor, S., & Hevner, A. R. (2013). MISQ 37(2), 337–355. DOI 10.25300/MISQ/2013/37.2.01 — [VER-P]
19. Hevner, A. R. (2007). *A Three Cycle View of DSR*. SJIS 19(2). https://aisel.aisnet.org/sjis/vol19/iss2/4/ — [VER-P]
20. Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS. EJIS 25(1), 77–89. DOI 10.1057/ejis.2014.36 — [VER-P]
21. Barr, E. T., et al. (2015). *The Oracle Problem in Software Testing*. IEEE TSE 41(5), 507–525. DOI 10.1109/TSE.2014.2372785 — [VER-P]

10.5 Usability & validity

22. Brooke, J. (1996). SUS. DOI 10.1201/9781498710411-35 — [VER-P]
23. Bangor, A., Kortum, P. T., & Miller, J. T. (2008). IJHCI 24(6), 574–594. DOI 10.1080/10447310802205776 — [VER-P]
24. Bangor, A., Kortum, P., & Miller, J. (2009). *Determining What Individual SUS Scores Mean*. JUS 4(3), 114–123 — [VER-S]
25. Lewis, J. R. (2018). IJHCI 34(7), 577–590. DOI 10.1080/10447318.2018.1455307 — [VER-P]
26. Fiandhika, M. F., & Kelana, B. (2024). E3S Web Conf. 483, 03008. DOI 10.1051/e3sconf/202448303008 — [VER-P]
27. Lynn, M. R. (1986). DOI 10.1097/00006199-198611000-00017; Polit & Beck (2006). DOI 10.1002/nur.20147 — [VER-P]

10.6 Tools & software (re-verify [2ND-UNV] rows)

28. ISACA **COBIT 2019 Design Guide Tool Kit** — [VER-P] design/prioritisation only (ZIP `isaca.org/-/media/files/isacadp/feature/downloads/c/cobit-2019-design-toolkit_tkt_eng_0222.zip`; single `.xlsx`; DF1–DF10 implemented, DF11 excluded; protected `SUMPRODUCT` formulas; no 0–5 computation; single-assessment). **Free "COBIT 2019 Tool Kit"** (ZIP `…/isaca/resources/cobit-2019-toolkit.zip`; reference-only; includes *Introducing COBIT 2019: Major Differences with COBIT 5* — the official "no PAM" statement). Critiques: **Utomo et al. (2022)**, CommIT J. 16(2):129–141, DOI 10.21512/commit.v16i2.8172 ("rigid … protected formula"); **Amore et al. (2023)**, KnE Soc. Sci., DOI 10.18502/kss.v8i1.12636 ("limited prescriptive power").
29. ServiceNow IRM/GRC; RSA Archer; MetricStream; SAP GRC; OneTrust; LogicGate; Diligent HighBond; Hyperproof; Wolters Kluwer TeamMate; Thomson Reuters ONESOURCE — vendor pages — [2ND-UNV]
30. ControlWeave; A-COBIT-2019-Risk-Assessment; evaluador-ti — repositories — [2ND-UNV]
31. CMMI Cybermaturity Platform — [2ND-UNV]

10.7 Academic prototypes & Indonesian studies — re-verified at DOI level (§5.4); per-paper scoring details still [2ND-UNV]

32. Mandiangan (2023), UPN Veteran Jakarta (repo 28558) — [VER-S]; Abdullah, Putra & Kartika (2023), DOI 10.59581/jusiik-widyakarya.v1i4.1319 — [VER-S]; Anwar & Harits (2025), DOI 10.55903/jitu.v3i1.271 — [VER-S]; Noor (2021), ITTP Purwokerto (repo 6445) — [VER-S]; Pamuji et al. (2022), DOI 10.34010/jtk3ti.v8i2.8131 — [VER-S]; Kurniawan et al. (2025), DOI 10.33480/inti.v19i2.6389 (COBIT 5) — [VER-S]; Indrawati et al. (2023), DOI 10.15294/edukomputika.v10i1.59761 — [VER-S]. **Removed from novelty discussion:** "Cobvapps (UB)" ([NF] — not found; likely misattributed). 32b. **Per-level-% + ">85% to continue" gate precedents** (defensible operationalisation of the prototype's rule): Marcela & Fianty (2023), *Indonesian Journal of Computer Science* 12(4):1763–1766, DOI 10.33022/ijcs.v12i4.3353 — [VER-P]; Simatupang & Fianty (2023), *G-Tech* 7(4), DOI 10.33379/gtech.v7i4.3141 — [VER-P]. Both reproduce the ISO/IEC 33020 bands verbatim and stop the assessment when a level "does not reach 85%". 32c. **Indonesian adoption proxies** (see `04-regulasi-...`): Akbar & Saputra (2023), *Sebatik* 27(2):589–605, DOI 10.46984/sebatik.v27i2.2336 — [VER-P] (year corrected from 2025); Sari, Ginardi &

Indrawanti (2023), *J. Teknik ITS* 12(1):A9–A14, DOI 10.12962/j23373539.v12i1.100436 — [VER-P]; Afrianda, Wirani & Sucahyo (2024), *IJCS* 13(4), DOI 10.33022/ijcs.v13i4.4145 — [VER-P] (primary abstract says "13 processes … below maturity level 3"; the "Level 0" detail is secondary — flag it); Irzavika & Mahda (2024), *ICIMCIS* pp. 692–696, DOI 10.1109/ICIMCIS63449.2024.10957022 — [VER-P].

10.8 AI / RAG (optional section only)

33. Karakurt, E., & Akbulut, A. (2026). Appl. Sci. 16(1), 368. DOI 10.3390/app16010368 — [VER-S] (existence); specific figures [2ND-UNV]
34. Eulerich, M., Masli, A., Pickerd, J. S., & Wood, D. (2023). CAR 40(2), 981–1012. DOI 10.1111/1911-3846.12847 — [VER-P]
35. KPMG. (2025). *SOX Survey* — [VER-S] (existence); details via optro.ai [2ND-UNV]

10.9 Indonesian regulation — verified 2026-10-13 (supersedes the earlier pending note)

36. **PP No. 71/2019** (PSTE) — Pasal 22(1) verbatim audit-trail obligation; Pasal 103(2) revokes PP 82/2012. [VER-P] https://peraturan.bpk.go.id/Details/122030/pp-no-71-tahun-2019 (PDF full-text checked: "COBIT" = 0)
37. **Perpres No. 95/2018** (SPBE) — Pasal 1(2), Pasal 3(c) (Audit TIK), Pasal 55(5). "COBIT" = 0. [VER-P] https://peraturan.bpk.go.id/Details/96913/perpres-no-95-tahun-2018
38. **Perpres No. 132/2022** — *Arsitektur SPBE Nasional*; **not** an amendment to 95/2018. [VER-P] https://peraturan.bpk.go.id/Details/233483/perpres-no-132-tahun-2022
39. **Permenkominfo No. 16/2022** (Audit TIK) — Pasal 1(2) definition; "standar dan praktik-praktik terbaik" is from the Lampiran's definition of "kriteria". "COBIT" = 0. [VER-P] https://peraturan.bpk.go.id/Details/255601/permenkominfo-no-16-tahun-2022
40. **PermenPANRB No. 59/2020** — SPBE evaluation (4 domains / 8 aspects / 47 indicators / maturity 1–5); **revoked by PermenPANRB No. 8/2026**. [VER-P] https://peraturan.bpk.go.id/Details/150381/permen-pan-rb-no-59-tahun-2020 ; https://peraturan.bpk.go.id/Details/351330/permen-panrb-no-8-tahun-2026
41. **Pedoman Menteri PANRB No. 3/2024** — current SPBE-evaluation technical guide. [VER-P]
42. **PermenPANRB No. 5/2018** (revoked) · **POJK 38/2016** (revoked) · **SEOJK 21/2017** (revoked) — the only instruments that **name COBIT**, always optionally/referentially. [VER-P]

Bottom line: no Indonesian regulation *mandates* COBIT; the only explicit COBIT references are optional best-practice mentions, mostly in revoked instruments. This is **context** for the thesis, not the justification for the tool.

Appendix A — Prototype algorithm vs published CPM (one-glance)

Aspect	ISACA / published	Prototype	Verdict
Level construct	Capability per process; maturity per focus area	Capability per objective	OK
Scale	0–5	0–5 subjective, base 1	OK
Rating granularity	F/L/P/N **per activity per level**, judgment + evidence	Yes/Partial/No per activity → % → F/L/P/N	Simplified; disclose
Thresholds	>85 / 50–85 / 15–50 / <15	≤15 / ≤50 / ≤85 / >85	Match modulo boundaries
Aggregation	**Not mandated**; simple avg proposed for early maturity	Simple mean	Permitted-as-proposal; disclose

Gate	Published proposal: all activities L or F	selectable: `aggregate-85` (default) / `all-l-or-f` / `iso-staged`	☐ Configurable (2026-10-13)
N.A.	Judgment ("judgment call")	excluded; all-N.A. → **Insufficient Evidence**	☐ Handled (2026-10-13)
Evidence	Traceability required	evidenceIds field + **status workflow** (pending→reviewed→attached)	Partial — status flow added; linkage not hard-enforced
Cases (121)	1,202 activities / 231 count practices / 40 objectives / 5 domains	subset	Scope-limited

Appendix B — Open items (status as of 2026-10-13)

#	Item	Status
1	Re-verify every `[2ND-UNV]` tool/prototype row from the primary paper/vendor page	☐ **Mostly done** — academic prototypes re-verified at DOI/repository level (§5.4, §10.7). ☐ **Still open:** commercial GRC rows (§5.2) — Vanta/AuditBoard/TeamMate pages were blocked; re-verify before the thesis lists them
2	Verify IIA 14.6 and ITAF 1205 exact numbering/title	☐ **Done** (2026-10-13): IIA 14.6 = *Engagement Documentation*; ITAF 4th ed. 1205 = *Evidence* (§6.1, §10.3)
3	Verify Indonesian regulation status	☐ **Done** (§9.5, §10.9): no mandate for COBIT; PermenPANRB 8/2026 revoked 59/2020; Perpres 132/2022 is not an amendment
4	Decide + document the **gate rule** (all-L-or-F vs aggregate >85% vs ISO staged)	☐ **Implemented + decided (2026-10-13)** — all three rules selectable; **defended primary = `aggregate-85`**, with `all-l-or-f` and `iso-staged` reported as robustness checks. Rationale + thesis-ready paragraph in `03-decision-log.md` §5.3. Label it an *adaptation* in the write-up
5	Replace the "vs Excel" framing with a reference-calculation fixture	☐ **Done (2026-10-13)** — "vs Excel" removed from `02` §5.4, `05`, `03`, `README`, test banner, and UI labels; independent reference fixture `src/utils/__tests__/referenceCalculation.test.ts` built and green
6	Decide the seeded COBIT subset (objectives/activities) and its licensing	☐ **Open** — ISACA content is copyrighted; a small seeded subset + attribute must be decided and documented
7	Decide whether minimal persistence + history/comparison is **built** or **design-only**	☐ **Built (2026-10-13)** — localStorage registry + compact snapshots (`cobit_assessment_snapshots_v1`) + `CompareAssessments.tsx` baseline→candidate comparison. Scope it in the thesis as single-browser/client-side
8	Verify ITAF edition and IIA edition to cite	☐ **Done** — cite ITAF 4th ed. (2020; a 5th ed. exists in 2026), IIA 2024 Standards
9	Reconcile `04-regulasi-indonesia-cobit.md` against §9.2 rows 19–24	☐ **Done (2026-10-13)** — a 6-item correction banner at the top of `04` now covers rows 19–24 (incl. §9.2 #23, SEOJK 21/2017 §5.2/§7.6.1 naming COBIT)
10	Apply the §9.2 register rows 1–8, 13, 18 to `01`/`02`/`03`/`05`	☐ **Done (2026-10-13)** — capability names, all-L-or-F gate, attribution, Almeida→Souza Neto, Eulerich/KPMG, Mandiangan, Cobvapps removal, `[2ND-UNV]` flags all applied